

AHNJILI ZHUPARRIS, PHD

MEDIA RESEARCHER • AI ENGINEER • COMPUTATIONAL ARTIST

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Senior research scientist and AI engineer with a PhD in Medicine from Leiden University and end-to-end experience building and deploying NLP, computer vision and generative AI systems in production. I lead research on media narratives, cultural consumption and audience segmentation at Harmony Labs, and previously built machine-learning biomarkers from smartphone and wearable data for decentralised clinical trials. My work on surveillance and synthetic media has been funded by Mozilla and the Processing Foundation and exhibited at Ars Electronica.

EDUCATION

2020-2024	PhD in Medicine Leiden University, Netherlands
2016-2018	MSc Cognitive Neuroscience (Research) Radboud Universiteit Nijmegen, Netherlands
2012-2016	BSc Biomedical Sciences, Neuroscience, with Honours University of Edinburgh, United Kingdom

SKILLS & QUALIFICATIONS

PROGRAMMING	Python, R, PySpark, SQL, JavaScript, web scraping
ML & ANALYTICS	PyTorch, TensorFlow, scikit-learn, OpenCV, computer vision, data mining, statistical modelling
NLP	HuggingFace Transformers, retrieval-augmented generation (RAG), embeddings, topic modelling, sentiment analysis, network analysis
GENERATIVE AI	Latent diffusion (U-Net, VAE, cross-attention, CLIP), GANs, LLMs, Stable Diffusion, YOLO, RVC and speech synthesis, ElevenLabs
CLOUD	AWS, Azure, Google Cloud Platform, Heroku
LANGUAGES	English (native), Mandarin (B1), Dutch (B1)

WORK EXPERIENCE

Senior Research Scientist FEB 2025 - PRESENT
Harmony Labs

NLP & MEDIA RESEARCH	Lead statistically rigorous research on media narratives, cultural consumption and audience segmentation at scale, applying embedding-based clustering, topic detection and network analysis to extract actionable insight from large, messy datasets and build data-driven audience personas.
COMMUNICATION	Translate complex analyses into clear narratives, briefs and presentations for partners and decision-makers.

AI Engineer SEP 2023 - FEB 2025
RealFaceValue

DIFFUSION MODELS	Built and fine-tuned latent diffusion models (U-Net backbone, VAE encoder/decoder, CLIP-conditioned cross-attention) and GANs on AWS for automated image generation, with custom training loops and domain-specific loss functions.
PRODUCTION	Deployed generative models to production behind web and mobile applications, including inference optimisation and API integration with backend and frontend teams.

WORK EXPERIENCE (CONTINUED)

Lead Data Scientist

SEP 2019 - SEP 2023

Centre for Human Drug Research

TRIAL ANALYTICS	Led end-to-end data science for multiple clinical trials, from statistical analysis planning and study design through collection, analysis and reporting. Authored full research packages including protocols, SAPs and final study reports.
DIGITAL BIOMARKERS	Built classification and severity-estimation models on smartphone sensor data (text, audio, voice, location), wearables and electronic patient-reported outcomes for decentralised trial-at-home studies.
REVIEWS	Conducted systematic literature reviews on digital health technologies and disease classification to set research direction.

Freelance AI Engineer, Data Scientist & Science Communicator

JAN 2018 - PRESENT

Artificial Nouveau Studio

SOFTWARE	Built voice-cloning pipelines (RVC, speech synthesis), GAN and diffusion-based image generation systems, real-time browser-based computer vision tools (face detection, object tracking) and deepfake detection algorithms.
AUDIO & VOICE AI	Built multimodal systems including a framework that generates spatial audio from visual content (TTS, spatial positioning, real-time analysis) and real-time voice cloning driven by machine-listening algorithms. Exhibited at Ars Electronica and IMPAKT. Funded by the Processing Foundation and the Stimuleringsfonds.
SURVEY ANALYTICS	Built citizen-science mobile applications for Quantified Citizen and analysed large-scale Global Drug Survey data on drug use patterns and pharmacological effects.
TEACHING	Designed and delivered workshops on generative AI, computer vision and machine learning at universities across Europe, including HKU, ArtEZ, UAL and the Piet Zwart Institute. Panelist and juror on machine-learning ethics.

AWARDS & FELLOWSHIPS

2025-2026	Public AI Creative Fellowship MetaGov · USA Creative technologist fellow building narratives and tools around public AI governance.
2024	Processing Foundation Fellowship Processing Foundation · USA Screen-to-Soundscape, an open-source framework reimagining screen readers as immersive soundscapes.
2024	Digital Culture & Fresh Perspective Stimuleringsfonds Creatieve Industrie · NL
2020	Mozilla Creative Media Award Mozilla Foundation · USA Future Wake, using AI on fatal police encounter data in the US to predict future incidents.

SELECTED PUBLICATIONS

2024	ZhuParris, A. <i>Development of machine learning-derived mHealth composite biomarkers for trial@home clinical trials.</i> PhD thesis, Leiden University.
2024	Wevers, R. <i>Embodying data, shifting perspective: a conversation with Ahnjili ZhuParris on Future Wake.</i> In: <i>Doing Digital Migration Studies</i> , Amsterdam University Press.
2023	ZhuParris, A. et al. <i>A smartphone- and wearable-based biomarker for the estimation of unipolar depression severity.</i> <i>Scientific Reports</i> .
2023	ZhuParris, A. et al. <i>Treatment detection and MDS-UPDRS Part III estimation using finger tapping tasks.</i> <i>Movement Disorders</i> .
2023	ZhuParris, A. et al. <i>Machine learning techniques for developing remotely monitored central nervous system biomarkers using wearable sensors: a narrative literature review.</i> <i>Sensors</i> 23, 5243.

REFEREES

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